

Contracting with Simplified Models

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1 Introduction

Across many economic environments, a principal will often contract an informed agent to provide data analysis that guides her decisionmaking. A CEO may enlist a management consultant to guide workforce restructuring using data from other firms’ experiences. An investor may hire a portfolio manager to inform her investment decisions using historical returns. A voter may ask a politician to justify a proposed transit subsidy by pointing to evidence from a pilot program. Or, a graduate student may consult an AI agent for help with a mathematical proof, drawing on academic literature the model was trained on.

An agent seldom provides the principal with raw data. Instead, he presents a *model* of the data — a data-generating process (DGP) that maps the principal’s choices onto outcomes of interest. Such a model may describe how a firm’s profits vary with the intensity of layoffs, or how investment returns depend on portfolio choices. Providing models rather than raw data is often desirable for a principal who seeks a clear recommendation, may not know how to process data, or prefers not to waste organizational resources on analysis. Meanwhile, the agent may contribute domain expertise or access to proprietary data that the principal cannot obtain herself.

However, learning via a model instead of data itself may constrain a principal’s ability to make an informed decision. The model an agent presents may be *oversimplified*, neglecting key tradeoffs or considerations that the principal finds important. A principal may also be uncertain about the *data* on which a model was estimated — such as whether the agent drew on sources highly relevant to the principal’s decision, or incorporated information from extraneous ones. Moreover, the model an agent supplies may be affected by whether the principal can gather information from other agents, and thus the broader “market for models.”

This paper theoretically studies how contracting with models and data is shaped by model oversimplification, uncertainty about data sources, and competition between model providers. A receiver (principal) chooses among actions and contracts a sender (agent) to provide information about a pre-specified action. The sender gathers relevant data and supplies a *model*: a DGP mapping actions to outcomes. The model may misrepresent tradeoffs the receiver believes are important, and the receiver does not see the data it was estimated on.

I address four main questions. First, if the sender estimates an oversimplified model, how does the receiver learn about the true DGP, and how does this depend on beliefs about the *data* it was estimated on? In particular, how does learning differ when the sender “locally samples” data in a neighborhood of a receiver’s specified action, as opposed to including information about

“distantly sampled” actions that are far from it? Second, does the sender sample locally or distantly, and how does this depend on the class of models he can supply? Third, does a sender have an incentive to provide oversimplified models? Finally, how are these distortions shaped by *competition* among senders — including market power, overhead costs, and the receiver’s ability to consult multiple senders?

In the model, the receiver chooses an action x to maximize an expected outcome y . She believes outcomes are generated by a DGP of the form $y = \alpha_0 + \alpha_1 x - \gamma x^2$ for $\gamma \geq 0$, capturing a tradeoff between a linear benefit and a convex cost with intensity γ . She specifies an action x_S for the sender to investigate. The sender then selects a sampling distribution over actions x with mean x_S . “Local” sampling corresponds to a low-variance distribution concentrated near x_S , while “distant” sampling corresponds to a high-variance distribution that places weight on actions farther from x_S . The sender draws actions x from this distribution and observes corresponding outcomes y generated by the DGP, producing data (y, x) . His sampling choice is unobserved by the receiver.

The sender then finds the model that best-fits a data. He considers a similar set of models as the receiver, except that the set of γ values he can consider is a subset of the receiver’s, limiting his ability to model *tradeoffs*. I allow for heterogeneous sender types who differ in the range of γ values they can model. Of particular interest is the case $\gamma = 0$, in which the sender is restricted to “linear regression” models. The receiver observes the sender’s model and updates her beliefs over the true DGP. The sender would like to maximize the receiver’s posterior expectation of y — because he is compensated based on her expected outcomes, seeks to maximize engagement, or prefers she hold optimistic beliefs.

Receiver Learning and Sender Sampling The first key result is that the sender’s modeling limitations interact with his sampling strategy to shape how the receiver learns. Under local sampling, the receiver’s set of plausible models is tangent to the sender’s model at $x = x_S$. Under distant sampling, this tangency disappears, and the set of plausible models shifts vertically. Models with higher γ — which are more concave and thus steeper on average — lie above the sender’s model at x_S , while less concave models with lower γ lie below. Because all plausible models must fit the same data, they share the same sample mean, implying that steeper models lie above the sender’s model near x_S and below it further away.

This leads to a second result: when the sender’s models tend to *understate* tradeoffs — i.e., restrict attention to models that are too shallow — he prefers distant sampling. In this case, any additional models the receiver entertains are steeper than the one the sender provides. If the receiver believes the sender used distant sampling, the maxima of these steeper models exceed that of the sender’s model. Thus, the sender’s model effectively acts as a lower bound for the maximal outcome y . By contrast, when the sender’s models tend to *overstate* tradeoffs, he prefers local sampling. In this case, any additional models the receiver entertains are *shallower* than the one the sender provides. The relevant (shallower) models are tangent at x_S and, because they are shallower, lie above the sender’s model at their maxima. Under distant sampling, these same models

would lie below the sender’s model. A key corollary is that if the sender is restricted to *linear regression* models ($\gamma = 0$), he always prefers distant sampling. This highlights a tension between linear regressions as “best local approximations” of nonlinear processes and their strategic use by model providers to help strategically make a receiver more optimistic about an outcome.

Third, this interaction between model misspecification and sampling has implications for receiver welfare. Under local sampling, the receiver’s posterior models are tightly concentrated in a neighborhood of x_S , leading to relatively low dispersion in outcomes at the receiver’s optimal choice. Under distant sampling, the vertical dispersion across plausible models increases both at x_S and at the receiver’s optimum. If the receiver values forecast accuracy, she benefits less from distant than from local sampling. Moreover, as sampling becomes increasingly distant — i.e., as the variance of the x distribution grows — the variance of outcomes increases without bound. Thus, the receiver learns the true DGP more precisely when the sender overstates tradeoffs, which is precisely when he has an incentive to *locally* sample.

Gains from Oversimplification The fourth result is that a sender may gain from model oversimplification. I consider a “sophisticated” sender who can fit any γ in the receiver’s feasible set, and ask when he benefits from the presence of “simple” senders who are restricted to a narrower range of γ values. A necessary condition for the sophisticated sender to gain from the simple sender’s existence is that the boundary of the simple sender’s γ -set intersects the interior of the sophisticated sender’s set. In this case, upon observing a model, the receiver assigns positive probability to either having received the correct model from the sophisticated sender or a misspecified model from a simple sender. In the latter case, the simple sender’s sampling induces the receiver to entertain alternative models that lie above the reported model in expectation, so that the reported model acts as a lower bound on expected payoffs. The presence of simple senders therefore raises the receiver’s posterior expected utility.

I use this result to study how a “type-space designer” can choose the distribution of sender types to maximize payoffs. For example, an AI firm may deploy models of varying sophistication to a user, or a consulting firm may assign teams with different modeling capabilities to a client. I show that an optimal design features a rich set of sender types, each only able to fit models with a single γ value. The key implication is that organizations may benefit from selecting a sender who has a fixed (but completely arbitrary) way of modeling the intensity of the payoff γ .

Sender Competition The final set of results concerns competition between multiple senders, who face costs of providing information (in the form of overhead fees) and thus charge prices for their services. I consider a heterogeneous set of receivers who differ in the actions they would like to receive information about. Senders position themselves throughout the action space and specialize in providing information about different actions.

The first observation is that if the action space is bounded, senders who specialize in providing information about extreme actions must necessarily sample “less distantly,” as the variance of the

distribution of x values they can collect is limited by their proximity to the boundary. This means that low γ senders who tend to understate tradeoffs will want to concentrate themselves in the center of the action space. High γ senders who overstate tradeoffs have no preferences on their location in the action space. However, senders who prefer distant sampling are made worse off if receivers believe they locally sample and vice-versa. This means distant and local samplers will never want to locate themselves in similar regions.

As a result, competition introduces *spatial sorting* across sender types. Senders who understate tradeoffs locate themselves in the center of an action space and distantly sample, while those who overstate tradeoffs locate closer to the boundary. If receivers value forecast error — and thus prefer local sampling — those who would prefer information about the interior of the action space also face a tradeoff between learning accurately about actions that are less valuable (in the middle of the action space) and less accurately about actions that are more valuable (near the boundary of the action space).

This tradeoff vanishes if overhead costs are sufficiently low and there are a large number of firms. In this case, a “market for second opinions” emerges, where a receiver contracts from two senders and combines their information to arrive at the true data-generating-process. Intuitively, this occurs precisely when it is cheap enough for a receiver to go to two senders — meaning costs must be low and that there is sufficient sender competition to encourage collecting two signals. I show that while this erases the potential gains for a “sophisticated” sender from the existence of “simple” sender types, it increases gains for a “simple” sender type by sharpening the range of plausible models the receiver entertains.

Section 2 lays out the basic model, and I characterize learning in section 3. Section 4 studies gains from oversimplifying models and the design of an optimal type space for a sophisticated sender. Section 5 investigates how these distortions become change when considering competition between multiple senders. Section 6 concludes. The remainder of this section gives an overview of the literature.

Literature

2 Benchmark Model

We begin by laying out a game between a single receiver R (she) and a single sender S (he). R decides whether she would like to contract the services of a sender S . She specifies a choice for S to investigate. S collects data about this choice and provides a R a simple model describing that data. R 's ability to commit S to a data sampling strategy or more complex models is limited. Taking this into account, R utilizes the information provided in S 's model to make a decision. We abstract from prices and competition, but return to study these questions in Section 5.

2.1 Receiver and Models

A receiver R faces a set of choices X , where X is a closed and convex subset of $\mathbb{R}$. Each of these choices maps to a distribution of outcomes $y \in \mathbb{R}$. Ex-ante, R does not know the distribution of outcomes. However, she believes that the distribution of y given x can be described by a data-generating process $y = f_R(x)$ for a function $f_R : X \rightarrow \mathbb{R}$. We refer to f_R as a **model**. We denote $\mathcal{F}$ as the set of *models* f_R that R entertains. $\mathcal{F}$ is closed and convex. We make the following assumptions on $\mathcal{F}$.

Assumption 1. *For each $f_R \in \mathcal{F}$, we have*

$$f_R(x) = \alpha_0 + \alpha_1 x - \gamma x^2$$

for $(\alpha_0, \alpha_1) \in \mathbb{R}^2$ and $\gamma \in \mathbb{R}^+$.

Assumption 1 says that each of R 's functions is concave, and can in particular be expressed as an affine function less a convex, quadratic loss. Notably, since $-\gamma x^2$ is concave, the functional form can also represent the consideration of diminishing returns in consumer utility, production, or any other concave function in shaping an outcome. We assume a quadratic form for tractability but relax this in section [sectionhere].

As a shorthand, we will denote the affine component of f_R as: $\ell(x; \alpha_0, \alpha_1) = \alpha_0 + \alpha_1 x$. We define as $\mathcal{L}$ the subset of functions $\mathcal{F}$ that are affine:

$$\mathcal{L} = \{f_S \in \mathcal{F} : f_S = \ell(x; \alpha_0, \alpha_1)\}$$

R has a prior p over the set of models $\mathcal{F}$. We make the following assumptions over the prior for tractability:

Assumption 2. *There exist densities p_α with support on $\mathbb{R}^2$ and p_γ with support on $[0, \bar{\gamma}]$ for $\bar{\gamma} > 0$ such that if $f_r(x) = \alpha_0 + \alpha_1 x - \gamma x^2$:*

$$p(f_r) = p_\alpha(\alpha_0, \alpha_1) \cdot p_\gamma(\gamma).$$

Moreover, $p_\alpha(\alpha_0, \alpha_1) = p_\alpha(\alpha'_0, \alpha_1)$ for all $\alpha_0, \alpha'_0 \in \mathbb{R}$.

This assumption says that the prior over the potential values of Γ is independent of the affine component $\ell(x; \alpha_0, \alpha_1)$, and that R places the same prior mass on vertical shifts of each model.

Given a belief q over models, R 's expected utility is given by:

$$U_q(x) = \mathbb{E}_q[f_R(x)].$$

Her goal is to choose x to maximize $U_q(x)$. The receiver values the expected value of the outcome y given her choice x . If R is an investor, she cares about how much money she can make by investing in a choice x . If R is a CEO restructuring her firm via layoffs, she cares about her decision's impact

on profits. If R is a voter, she cares about the impact of a policy x on her wellbeing. We utilize these simpler preferences in our baseline model, but section [sectionhere] extends them to evaluate settings where R also values *forecast accuracy*, and thus knowing the correct model of the world.

2.2 Sender and Data

The receiver R contracts a sender S to provide her with information about the true data-generating process. The sender has a private type affecting the simplicity of the models she considers. The order of events is as follows.

1. Nature draws S 's type.
2. R pre-specifies a choice $x_S \in X$.
3. S collects data about the choice x_S .
4. S provides a model to R that fits the data he collected. R updates her beliefs about the true data-generating process based on her beliefs about S 's type and the model.
5. R makes a decision x^* . S receives utility. Uncertainty resolves, and R receives her utility.

We now provide more detail about each of these components.

Sender's Type A sender's type affects the set of models — and hence information — he is capable of providing to R . The key is that S is limited in how he can model the non-affine component of f_R . To this end, we denote the type space by $\Theta \subseteq \mathcal{P}[0, \bar{\gamma}]$ — i.e. each type $\theta \in \Theta$ corresponds to a subset of $[0, \bar{\gamma}]$. The sender's type is drawn from a distribution p_θ over Θ , which is known to R . Θ has the following properties.

Assumption 3. For each $\theta \in \Theta$, $\theta \subseteq [0, \bar{\gamma}]$ is closed and convex. Moreover, for each $\theta \neq \theta'$, $\theta \cap \theta' = \emptyset$.

Assumption 3 states that each type corresponds to an interval within $[0, \bar{\gamma}]$. This interval may be degenerate. We assume the intervals do not intersect for expositional purposes, but the case where they do not intersect is of particular interest, and we study it in section [sectionhere].

A type θ sender is able to provide R a model in the set F_θ , where

$$F_\theta := \{f_R \in \mathcal{F} : f_R = \beta_0 + \beta_1 x - \delta x^2, \delta \in \theta, (\beta_0, \beta_1) \in \mathbb{R}^2\}.$$

Intuitively, θ represents the *degree* to which a sender can model the *non-linear* component of a data-generating-process f_R . For example, suppose there is a single type $\theta = [0, \bar{\gamma}]$. Then, S is able to provide R any of the models in $\mathcal{F}_R$. Of particular interest is the case where $\theta = \{0\}$, in which case S can only provide information to R in the form of a “linear regression.”

Sender's Data Given a pre-specified choice x_S , S chooses a data sampling strategy μ_S , where μ_S is a distribution defined over X . Intuitively, this means that S is choosing a distribution of choices in x to collect data on. R does not observe S 's choice of μ_S , meaning data-collection suffers from a *lack of verifiability and commitment*.

Assumption 4. μ_S is parametrized by $\mu_S(x_S, \sigma) = x_S + \sigma \cdot \epsilon$, where ϵ is mean-zero, symmetric, and $0 < \text{Var}(\epsilon) < \infty$. Moreover, $\sigma \in [\underline{\sigma}, \bar{\sigma}]$ for $\underline{\sigma} > 0$ arbitrarily close to 0.

Assumption 4 provides a convenient parametrization of μ_S as a distribution centered around x_S . σ as a parameter effectively controls the spread of the distribution. The constraint that $\sigma > 0$ is a normality assumption made to ensure learning is structured in equilibrium. We will refer to σ as a choice variable for S , and denote σ_θ^* the equilibrium choice of σ for a type θ sender.

Given this sampling strategy, S directly observes the joint distribution of (y, x) , subject to observational noise. Formally, if the true data-generating-process is f_R^* , she observes a joint distribution $D(y, x)$, where the marginal distribution of x is given by $D_x = \mu_S$ and the conditional distribution of y given x is given by $D_y(y|x) = f_R^*(x) + \eta$, where η is a mean-zero noise term with unknown variance. This is equivalent to S observing an infinite number of i.i.d. draws from the true data-generating-process.

Sender's Information With the sender's type and sampling strategy, in hand, we define the model S provides to R . We say that a model m **best-fits the data** with respect to θ and a data-sampling strategy μ_S if:

$$f_S = \arg \min_{f \in F_\theta} \mathbb{E}_{D(\mu_S)} [(y - f(x))^2].$$

Our key assumption is that S provides R the model f_S that best-fits the data, given his type θ and μ_S . The existence of f_S is given by the Hilbert Projection Theorem. For example, if $F_\theta = \mathcal{F}$, then f_S will represent the true data-generating process. Or, if $F_\theta = \mathcal{L}$, S provides the best-fitting *linear regression model* to the data. Crucially, both θ and μ_S are unobserved by R , although R may be able to infer these objects in equilibrium.

Sender's Preferences We assume type θ sender has a prior p^θ over the models in $\mathcal{F}_\theta$. We assume that p^θ can be decomposed into $p_\beta^\theta(\beta_0, \beta_1)$ and $p_\delta^\theta(\gamma)$ and accordingly satisfies Assumption 2. Suppose S provides a model $f_S \in \mathcal{F}_\theta$. Let x^* be the receiver's choice of $x \in X$ conditional on her posterior beliefs $q|f_S$. The sender's utility conditional on f_S is given by:

$$U_S(x^*) = \mathbb{E}_{q|f_S} [f_R(x^*)].$$

That is, conditional on providing a model f_S , S would like to maximize R 's expected utility. These preferences lend themselves to several interpretations. For example:

- S is a portfolio manager who receives a fixed share of R 's expected earnings. Then, he cares about the expected profits R can realize (and thus his payment).
- S is a consulting firm whose services a CEO R contracts. S may care about the expected gain in profits R receives either directly (e.g. because his "bonus" is proportional to R 's gain in profits) or indirectly (e.g. because his reputation is tied to his ability to realize profits for those who contract his services).
- S is an office-motivated politician who provides information to a median voter R . The higher the median voter thinks y is, the more likely she is to turn out. If re-elected, the politician will implement R 's preferred policy x^* .

2.3 Comments

The model assumes that R cannot verify information on two fronts: *data* and *functional form*. The lack of commitment to a specific dataset underscores a key feature of contracting with models. R knows S has access to useful data, but does not know exactly what that data looks like or how it is being used to provide her with a model. The clearest example is one where S 's data is exclusive and proprietary — R cannot observe these data directly, and must take into account S 's incentives and the information she is provided to make inferences about what the data might be. Portfolio managers and consulting firms may have access to past data on returns or cases they have worked on which they are unable to disclose directly, or politicians may have access to confidential data on past policy experiments collected by the public service. Another example is an AI model which detects high-level data patterns in a "black-boxed" way that cannot be elucidated to a user (or even its designers).

While the receiver R may be unable to access these data for legal reasons, she also be unable (or unwilling) to expend organizational time and resources in collecting, understanding, and synthesizing complex data. Instead, she delegates the collection of data and its synthesis into a model to the sender, expediting her decision-making process in the interest of convenience, but potentially at the cost of information.

This is related to the second dimension of nonverifiability, which is that S may not necessarily fit the true data-generating-process in the way R would like. The sender's type constrains how he can model the true data-generating-process, which may be affected by technological feasibility or a difference in thinking about a problem the same way R does. For example, suppose S is a climate scientist and R is an economist. S provides climate modeling to help R assess the impact of a climate shock on the demand for electric vehicles. Even if S understands the concept of demand, there may be technical constraints on the sorts of parameters his advanced climate models can provide to R . In particular, because $-\gamma x^2$ represents a convex loss, model simplicity particularly constrains S 's ability to model *tradeoffs*. Thus, S may be forced to provide a model that understates or overstates the degree of a tradeoff.

Many of the results will involve thinking about what happens when S can only provide linear regression models to R . This is motivated by the widespread usage of linear regression models in the business world, economics, and data-driven social science. We will be interested in understanding not only how we can learn about non-linear processes through linear models, but also how the use of linear models coupled with data sampling strategies can be used to manipulate beliefs about the effects of a policy or intervention. Moreover, we will contrast how the behavior of senders constrained to provide linear models may in particular differ from the behavior of senders constrained to provide other simplified (but nonlinear) models.

For the majority of the analysis, we focus on settings where the nonlinear component of f_R is quadratic. While this is done for tractability, section [sectionhere] discusses the sorts of assumptions necessary to generalize the logic to more general functional forms. Most of the results also generalize to multidimensional settings, and many of the proofs are identical, which we discuss in the same section.

3 Learning from Simple Models

3.1 Receiver's Learning and Behavior

We begin by characterizing receiver learning as a function of data-sampling strategies and S 's type.

Receiver's Updating Suppose S provided a model of the form $y = \beta_0 + \beta_1 x - \delta x^2$. What models would R entertain? First, note that due to Assumption 3, R is able to infer θ from δ . We define $\phi(f_S; \theta, \mu_S)$ as the set of functions that could rationalize the data, given f_S best-fits with respect to F_S and μ_S :

$$\phi(f_S; \theta, \mu_S) = \left\{ f_R \in \mathcal{F} : f_S = \arg \min_{f \in \theta} \mathbb{E}_{\mu_S}[(f_R(x) - f(x))^2] \right\}.$$

If $f_R \in \phi(f_S; \theta, \mu_S)$ We say that f_R **fits the data** with respect to f_S , θ , and μ_S if $f_R \in \phi(f_S; \theta, \mu_S)$. Since S 's type does not constrain β_0, β_1 , we necessarily have that for each $f_R \in \phi(f_S; \theta, \mu_S)$:

$$\frac{\partial}{\partial \beta_i} \mathbb{E}_{\mu_S}[(f_R(x) - f(x))^2] = 0,$$

which yields the following moment conditions familiar from ordinary-least-squares (OLS):

$$\mathbb{E}_{\mu_S}[f_R(x) - \beta_0 - \beta_1 x - \delta x^2] = 0 \tag{1}$$

$$\mathbb{E}_{\mu_S}[x(f_R(x) - \beta_0 - \beta_1 x - \delta x^2)] = 0 \tag{2}$$

Equation (1) is the moment condition with respect to the intercept, and states that on average, the difference between f_R and f_S must average to zero. Equation (2) can be combined with equation

(1) to read that the covariance between x and $f_R(x) - \beta_0 - \beta_1 x - \delta x^2$ is zero. Both expectations are taken with respect to μ_S , which is S 's sampling strategy.

Additionally, we also have a moment condition corresponding to δ :

$$\mathbb{E}_{\mu_S}[x^2(f_R(x) - \beta_0 - \beta_1 x - \delta x^2)] \quad (3)$$

which may be zero or nonzero, depending on S 's type.

The following proposition characterizes the set of functions that R entertains given f_S , θ , and μ_S .

Proposition 1. *Suppose $f_S = \beta_0 + \beta_1 x - \delta x^2$ and $\mu_S = \mu_S(x_S, \sigma)$. Then, there exists $\theta \ni \delta$ such that for all $f_R \in \phi(f_S; \theta, \mu_S)$, we have:*

$$f_R(x) = \beta_0 + \beta_1 x - \gamma(x - x_S)^2 + (\gamma - \delta)\sigma^2.$$

The set of models that R considers takes a simple form. There is a baseline “linear regression” component of each f_R that is the same as f_S : $\beta_0 + \beta_1 x$. There is then a quadratic component, centered around x_S (where γ is potentially unknown). Finally, there is a constant term that depends on γ , δ , and σ^2 . Notably, if $\theta = [0, \bar{\gamma}]$, ϕ will be a singleton containing the true DGP. Otherwise, there may be multiple models that rationalize the data.

The intuition can be described as follows. First, we utilize a result known as “Stein’s Lemma” (Landsman and Nešlehová, 2008) to show that (2) can be written as

$$\sigma^2 \mathbb{E}_\epsilon[f'_R(x_S + \sigma \cdot \epsilon) - f'_S(x_S + \sigma \cdot \epsilon)] = 0 \quad (4)$$

This says that the derivative of f_R and f_S must be the same “on average” over μ_S . As a result, as $\sigma \rightarrow 0$, for each $f_R \in \phi(f_S; \theta, \mu_S)$, the derivative of f_R and f_S converges at the point x_S . Using (1), we also have that as $\sigma \rightarrow 0$, the value of each f_R and f_S must converge at f_S . Thus, all $f_R \in \phi(f_S; \theta, \mu_S)$ are (approximately) tangent to f_S at x_S .

This intuition is shown in the left panel of Figure 1 below. The solid line shows f_S , which has a curvature δ . However, given the sender’s type θ may not account for all potential values that γ could take, there are other models f_R which could have generated the data, which are in turn reflected in the dashed blue lines. Crucially, all these lines are tangent to f_S at x_S .

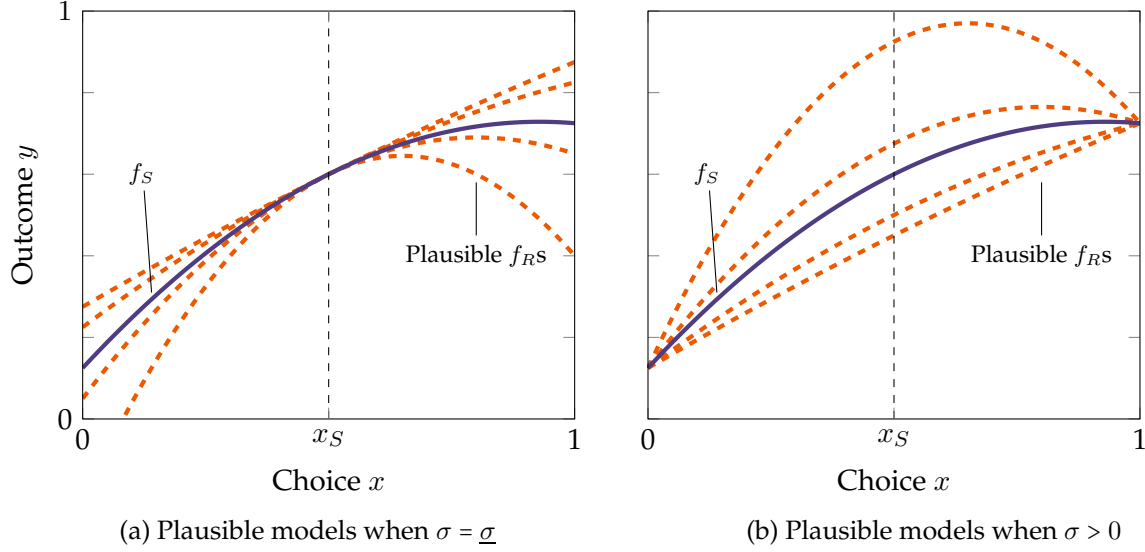


Figure 1: Plausible receiver models f_R as function of sender model f_S and sampling strategy $\mu_S(x_S, \sigma)$

The set of feasible models is different for $\sigma > 0$. In particular, if $\sigma' > \sigma$, then the distribution $\mu_S(x_S, \sigma')$ is a mean-preserving spread of $\mu_S(x_S, \sigma)$. Since f_R and f_S are both quadratic, their derivatives are linear, meaning (4) is invariant to mean preserving spreads (and thus does not depend on σ). Thus, each plausible f_R at σ is also plausible at σ' and vice-versa, up to a constant. The value of this constant depends on the curvature of f_R relative to f_S . Because each f_R is concave, its expectation over $\mu_S(x_S, \sigma)$ is decreasing in σ . If a given f_R is “more concave” than f_S — i.e. if $\gamma > \delta$, this f_R must shift up by a constant relative to f_S . If a given f_R is “less concave” than f_S — i.e. if $\gamma < \delta$, f_R must shift down by a constant relative to f_S . This is shown in the right panel of Figure 1. For higher σ , the plausible f_{RS} that are “more linear” lie below f_S in the sampling range. The f_{RS} that are “more concave” lie above f_S .

Receiver’s Decisionmaking The receiver’s learning takes a relatively simple form: regardless of the limitation on S ’s ability, R always learns what the data-generating-process looks like “up to the first derivative.” Moreover, as σ changes — and thus S ’s sampling strategy changes — the set of plausible f_{RS} remains identical, except that they shift up or down by a constant. This means that R ’s optimal *action* depends only her expectation of γ (given δ and θ). Given f_S , the expected value of the problem also has a simple expression. We denote For ease of notation, we will denote $\underline{\delta}_\theta = \min\{\theta\}$ and $\bar{\delta}_\theta = \max\{\theta\}$

Corollary 1. *R ’s optimal action is given by:*

$$x^* = x_S + \frac{\beta_1}{2\mathbb{E}_\gamma[\gamma|\theta, \delta]}.$$

R 's expected value from this action is given by

$$y^* = \beta_0 + \beta_1 x_S + \frac{\beta_1^2}{4\mathbb{E}_\gamma[\gamma|\theta, \delta]} + (\mathbb{E}_\gamma[\gamma|\theta, \delta] - \delta)\sigma^2.$$

The value of $\mathbb{E}_\gamma[\gamma|\theta, \delta]$ is also relatively simple.

- If (3) is equal to 0 at δ , then $\mathbb{E}_\gamma[\gamma|\theta, \delta] = \delta$.
- If (3) is ≤ 0 , then it must be that $\delta = \bar{\delta}_\theta$, and $\mathbb{E}_\gamma[\gamma|\theta, \delta] = \mathbb{E}[\gamma|\gamma \geq \delta]$.
- Likewise, if (3) is ≥ 0 , then it must be that $\delta = \underline{\delta}$, and $\mathbb{E}_\gamma[\gamma|\theta, \delta] = \mathbb{E}[\gamma|\gamma \leq \delta]$.

3.2 Sender's Data Sampling and Model Provision

We use these insights to derive the first result of the paper, which shows how S chooses a sampling strategy (i.e. σ) as a function of her type. We will use the following terminology to distinguish different types of sampling.

- **Local sampling:** $\sigma = \underline{\sigma}$. In this case, μ_S samples x values in a small neighborhood of x_S , i.e. is collecting information "locally relevant" to x_S .
- **Global sampling:** $\sigma = \bar{\sigma}$. In this case, μ_S samples x values that are farther and farther away from x_S , i.e. including information that potentially "irrelevant" (or at least spatially far) from x_S .

We also introduce the following notation

$$\begin{aligned}\bar{\gamma}^\theta &= \mathbb{E}_\gamma[\gamma|\gamma \geq \bar{\delta}_\theta] \\ \underline{\gamma}^\theta &= \mathbb{E}_\gamma[\gamma|\gamma \leq \underline{\delta}_\theta]\end{aligned}$$

These represent R 's expectation of γ , given noting that $\mathbb{E}_\gamma$ is taken with respect to R 's beliefs.

We now state our result on sender sampling.

Theorem 1. Suppose $\theta = [\underline{\delta}_\theta, \bar{\delta}_\theta]$. Then, the optimal degree of sampling for a type θ sender σ_θ^* is equal to either $\bar{\sigma}$ or $\underline{\sigma}$. In particular, S distantly samples if and only if

$$p_\gamma^\theta(\bar{\delta}_\theta)[\bar{\gamma}^\theta - \bar{\delta}] \geq p_\gamma^\theta(\underline{\delta}_\theta)[\underline{\delta} - \underline{\gamma}^\theta],$$

in which case $\sigma_\theta^* = \bar{\sigma}$. Otherwise, $\sigma_\theta^* = \underline{\sigma}$, and S locally samples.

Theorem 1 can be interpreted as follows. If $\delta \in (\underline{\delta}_\theta, \bar{\delta}_\theta)$, then R believes S 's model is correct. However, if $\delta = \bar{\delta}_\theta$, R will consider models with $\gamma \geq \bar{\delta}_\theta$. In this case, S 's tends to estimate models that understate the magnitude of a potential tradeoff. This creates an incentive for him to estimate his model on information that is potentially irrelevant to learning about the DGP in a neighborhood of x_S .

However, when $\delta = \underline{\delta}_\theta$, the opposite happens: R considers models with $\gamma \leq \underline{\delta}_\theta$, i.e. “flatter” than those that S provided. This gives S an incentive to estimate the DGP “locally” in a neighborhood of x_S . If a type θ S tends to consider models with only high values of γ — i.e. to “overstate” the sharpness of a tradeoff — then he does have an incentive to locally sample choices in a neighborhood of x_S .

The intuition is most easily seen in Figure 2 below. The left panels (a) and (c) look at the case where f_S is such that $\delta = \underline{\delta}_\theta$, and thus where R entertains models that are “less steep” than f_S . The right panels (b) and (d) look at the case where f_S is such that $\delta = \bar{\delta}_\theta$, and R entertains steeper models (with higher γ).

In panels (a) and (b), we first look at the case where $\sigma = \underline{\sigma}$ and S locally samples around x_S . In both cases, the set of plausible models f_R is approximately tangent to f_S at x_S . However, in panel (a), because these f_R s are “shallower” (i.e. have lower γ values than $\underline{\delta}_\theta$), each function lies *above* f_S . By contrast, in panel (b), each f_R is “steeper” than f_S (i.e. has $\gamma \geq \bar{\delta}_\theta$), and thus lie *below* f_S .

This logic flips when looking at panels (c) and (d). Recall that the distribution of x values when $\sigma = \bar{\sigma}$ is a mean-preserving-spread of that when $\sigma = \underline{\sigma}$. To rationalize the data, functions with higher γ values that are “more concave” must shift up relative to shallower ones. In panel (c), this means that the shallower f_R functions now lie *below* f_S . In panel (d), the steeper f_R functions now lie *above* f_S .

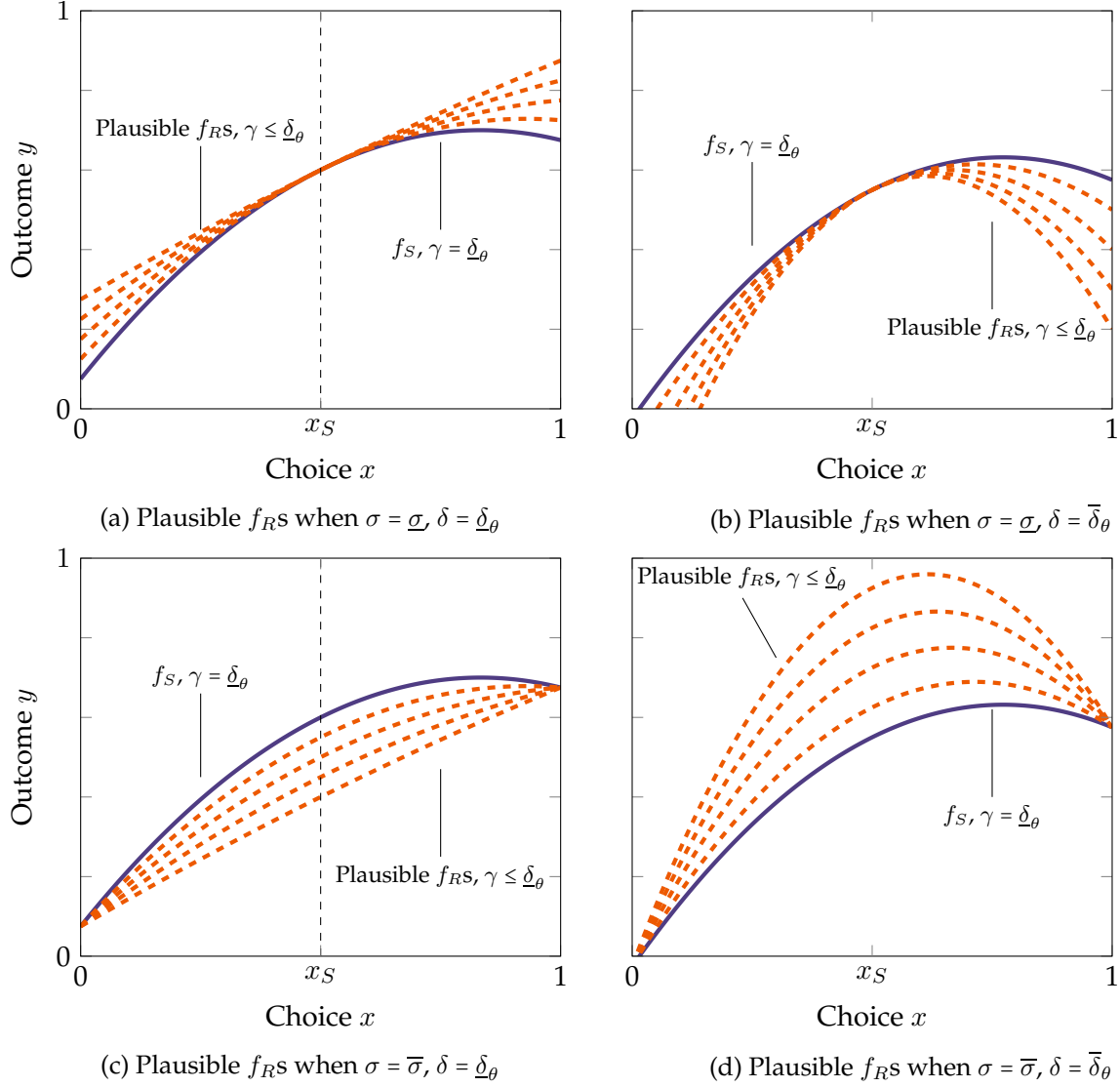


Figure 2: Models R entertains when $\delta = \underline{\delta}_\theta$ vs. $\delta = \bar{\delta}$, local sampling ($\sigma = \underline{\sigma}$) vs. distant sampling ($\sigma = \bar{\sigma}$)

Thus, whether $\sigma_\theta^* = \bar{\sigma}$ or $\underline{\sigma}$ depends on which scenario is more likely for S . If he is likely to end up in situations where the set of functions R will see is “shallower” than the “shallowest” function his type can model, he will prefer local sampling, as this effectively makes his model a lower bound for the set of models R entertains. If he is likely to end up in situations where the set of functions R will see is “steeper,” he will prefer distant sampling.

In an extreme case, suppose $\theta = [\underline{\delta}_\theta, \bar{\gamma}]$. In this case, S can model the steepest tradeoffs R will entertain, but not shallower ones. Thus, he will always prefer local sampling. On the other hand, suppose $\theta = [0, \bar{\delta}_\theta]$. Then, any additional models that R entertains are “steeper” than f_S , so he has an incentive to distantly sample data and drive up R ’s expected value of these steeper problems.

Of particular interest is also the situation where $\theta = \{0\}$, which corresponds to a type of S who

can only model *linear regressions*. Using Theorem 1, not only does such a sender always have an incentive to sample distantly, but as $\sigma \rightarrow \infty$, his perception of R 's utility from his model also goes to ∞ .

Corollary 2. *Suppose $\theta = \{0\}$, so that S only sends linear models. As $\sigma \rightarrow \infty$, $V(x_S) \rightarrow \infty$. S 's perceptions about R 's value also go to ∞ .*

This logic is also illustrated in

3.3 Receiver's Choice of x_S and Value of Information

Value of Information In the absence of information, R relies solely on her prior to choose x^* .

$$V(x_S) = \int_{\Theta} \mathbb{E}_{\beta_1, \gamma} \left[\frac{\beta_1^2}{4\gamma} \right] Pr(\delta \in \theta) + \frac{\mathbb{E}_{\beta_1}[\beta_1^2]}{4\mathbb{E}[\gamma]} - \mathbb{E}[\beta_1]^2 \frac{1}{4\mathbb{E}[\gamma]} \quad (5)$$

3.4 Forecast Accuracy

4 Gains from Simplification

5 Competition

6 Conclusion

References

Landsman, Z. and J. Nešlehová (2008). Stein's lemma for elliptical random vectors. *Journal of Multivariate Analysis* 99(5), 912–927.